Cyber Security

28-Day Syllabus

Cyber Security Fundamentals & OS Basics

Cybersecurity Overview

- Cybersecurity Concepts, Threats, and Vulnerabilities
- Types of Cyber Attacks (Phishing, DoS, APT, Malware)
- The CIA Triad (Confidentiality, Integrity, Availability)
- Cybersecurity Frameworks (NIST, ISO 27001)

Introduction to Linux

- Linux OS Overview and Command Line Basics
- File Management, Permissions, Users/Groups
- Basic Linux Security Tools (iptables, Nmap, Wireshark)

Windows Basics

- Windows OS Overview and Architecture
- User Privileges, Registry, Event Logs
- Basic Windows Security Tools (Procomn, PsExec)

Networking & Security

Networking Basics

- OSI & TCP/IP Models, Protocols (IP, TCP, UDP, HTTP)
- Routers, Switches, Firewalls, IDS/IPS

Network Security Basics

- Firewalls: Types (Packet Filtering, Stateful, Proxy)
- IDS/IPS Concepts and Configurations

Cryptography & Wireless Security

- Symmetric vs Asymmetric Encryption
- Cryptographic Algorithms (AES, RSA, SHA-256)
- Wireless Network Security (WPA3, Rogue Access Points)

Vulnerability Assessment & Pen Testing

Vulnerability Scanning

- Vulnerability Management (CVE, CVSS)
- Tools: Nessus, OpenVAS, Qualys
- Vulnerability Assessment Process

Penetration Testing Basics

- Pen Test Phases: Reconnaissance, Scanning, Exploitation
- Ethical Hacking Guidelines
- Pen Test Tools: Nmap, Metasploit, Burp Suite

Incident Response & Malware Analysis

Incident Response

- Incident Response Lifecycle
- Building an Incident Response Plan

Malware Analysis

- Malware Types (Viruses, Trojans, Ransomware)
- Static and Dynamic Malware Analysis

Digital Forensics

- Chain of Custody, Evidence Collection
- Tools for Forensics (FTK Imager, EnCase)

Mobile, IoT & Cloud Security

Mobile Security Basics

- Mobile OS Security (Android, iOS)
- Mobile App Security Testing & Penetration Testing

IoT Security

- IoT Device Vulnerabilities and Security Challenges
- IoT Pen Testing

Cloud Security

- Cloud Models: IaaS, PaaS, SaaS
- Cloud Security Best Practices (IAM, MFA)

Threat Hunting & Defensive Security

Threat Hunting & Detection

- Threat Hunting Process and Tools (EDR, SIEM)
- Investigating Indicators of Compromise (IoC)

Defensive Security & Hardening

- Operating System Hardening (Linux/Windows)
- Network Security Hardening (VPNs, Firewalls)